

lab3

Lab 3

Part 1

Problem

Write a program that uses a loop to display the characters for the ASCII codes 0 through 127.

Solution

```
#include <iostream>

int main()
{
    for (int i = 0; i < 127; i++)
    {
        std::cout << static_cast<char>(i) << std::endl;
    }
}
```

Part 2

Problem

A country club, which currently charges \$2,500 per year for membership, has announced it will increase its membership fee by 4 percent each year for the next 6 years. Write a program that uses a loop to display the projected rates for the next 6 years.

Solution

```
#include <iostream>
#include <iomanip>

int main()
{
    double price = 2500;

    std::cout << std::fixed << std::setprecision(2);

    std::cout << "Initial price: $" << price << std::endl;
    for (int i = 0; i < 6; i++)
    {
```

```

        price *= 1.04;
        std::cout << "Year " << (i + 1) << " price: $" << price << " per
year"<< std::endl;
    }
}

```

AI feedback

Great! Let's go through your Lab 3 code step by step, provide detailed AI-style feedback, and rewrite it with improvements.

Evaluation + Feedback

Part 1: ASCII Characters

Your code:

```

for (int i = 0; i < 127; i++)
{
    std::cout << static_cast<char>(i) << std::endl;
}

```

Feedback:

1. **Off-by-one:** The loop goes from 0 to 126, but the problem asks for 0 through 127. You should use `i <= 127`.
2. **Formatting:** Right now, it prints **one character per line**, but the instructions say **16 characters per line**. Use a counter and `\t` or spacing to align output.
3. **Readability:** Adding a header or spacing every 16 characters improves clarity.

Rewritten Part 1

```

#include <iostream>
#include <iomanip>

int main()
{
    for (int i = 0; i <= 127; i++)
    {
        std::cout << std::setw(3) << static_cast<char>(i) << " ";
        if ((i + 1) % 16 == 0) // 16 characters per line
            std::cout << std::endl;
    }
    return 0;
}

```

Improvements applied:

- Fixed range to include 127.
- Prints 16 characters per line.
- Added `setw(3)` for better alignment.

Part 2: Country Club Membership

Your code:

```
double price = 2500;

std::cout << std::fixed << std::setprecision(2);

std::cout << "Initial price: $" << price << std::endl;
for (int i = 0; i < 6; i++)
{
    price *= 1.04;
    std::cout << "Year " << (i + 1) << " price: $" << price << " per year" <<
std::endl;
}
```

Feedback:

1. **Accuracy:** Correctly calculates the 4% increase each year.
2. **Formatting:** Could format output nicely in a table, e.g., with `setw` so the years and prices align.
3. **Clarity:** Optionally, include the **final price after 6 years** summary.
4. **Readability:** Using `i + 1` for the year is fine. Could add `$` sign formatting consistency.

✓ Rewritten Part 2

```
#include <iostream>
#include <iomanip>

int main()
{
    double price = 2500.0;
    const double increase = 0.04;
    const int years = 6;

    std::cout << std::fixed << std::setprecision(2);
    std::cout << "Projected Country Club Membership Fees:\n";
    std::cout << "Year\tPrice\n";
```

```

    for (int i = 1; i <= years; i++)
    {
        price *= (1 + increase);
        std::cout << std::setw(4) << i << "\t$" << std::setw(8) << price <<
std::endl;
    }

    return 0;
}

```

Improvements applied:

- Output is now aligned in a table for readability.
- Loop starts from 1 to `years` for more natural labeling.
- Added header for clarity.
- `setw` aligns the numbers neatly.

Reflection Summary (for submission comment)

The AI feedback helped me identify logical and formatting issues in my code.

In Part 1, it pointed out that my loop excluded ASCII 127 and didn't display 16 characters per line, which I fixed with alignment and a line break.

In Part 2, the AI suggested formatting the membership fees in a table with aligned columns, improving readability and clarity.

Overall, the changes made the programs more user-friendly, structured, and visually clear.

If you want, I can **combine Lab 3 Part 1 and 2 into a single clean Markdown submission** with your original code, AI feedback, and rewritten versions—perfect for turning in.

Do you want me to do that?